

Why do we use $n-1$ in statistics:-

→ when we calculate variance or standard deviation, we first compute mean from the same data.

→ This creates a restriction $\Rightarrow$ one value is no longer free

Ex: 5, 8, 10 (computing mean)

$\Rightarrow$ If mean = 7.66, If two values are known, the 3rd value is fixed

$\Rightarrow$ So only $n-1$ values are free to vary

$\therefore$ we divide by $n-1$ to get an unbiased estimate of Population Variance

Using Statsmodel ^{LR} [model.summary()]

i) finding whether there is ^{linear} correlation $X | Y$
ship b/w X & Y using hypothesis test [F-statistic]

ii) if it is a linear relation, finding the strength b/w X and Y on goodness of fit using R^2 score & adjusted

iii) $Y(X_1, X_2, X_3) \rightarrow$ analyzing how independent cols affecting the targeted val Y .

F-Statistic & Prob(F-Statistic) :- $\left\{ \begin{array}{l} \text{It is a test conducted} \\ \text{to check the relation b/w} \\ \text{X \& Y exists or not} \end{array} \right.$

The F-test for overall Significance is a statistical test used to determine whether a linear regression model statistically significant. [meaning it provides a better fit to the data than just using the mean of the dependent variable (Y)]

$\Rightarrow$ F-test regression checks whether your independent variables actually helps to explain the dependent variable.

$\Rightarrow$ If the F-test is significant, means your overall Reg model is significant

Steps involved in conducting an F-test :-

Step 1:- State the null and alternative hypothesis

→ Null hypothesis (H_0): All regression coefficients (except β_0) are equal to zero $\{ H_0 \rightarrow (\beta_1, \beta_2, \beta_3, \dots, \beta_k = 0) \}$
 → meaning that none of the independent variable contributes significantly to the explanation of the dependent variable's variation.

→ Alternative hypothesis (H_1): At least one regression coefficient is not equal to zero $\{ H_a \rightarrow \beta_1 \neq 0 \text{ or } \beta_2 \neq 0 \text{ or } \dots \}$
 → indicating that at least 1 independent variable contributes significantly to the explanation of the dependent variable's variation.

Step 2:- fit the L.R model to the data, estimating the regression coefficients (intercept & slopes).

Step 3:- Calculate the sum of Squares values

i) $TSS = \sum (y_i - \bar{y})^2$

iii) $ESS = TSS - RSS$

ii) $RSS = \sum (y_i - \hat{y})^2$

$df_1 = \text{degree of freedom 1}$

Step 4:- Compute the Mean Squares (MS) values

F-statistic

follows

F-distribution

obtained by

$$\frac{\frac{\lambda_1^2}{df_1}}{\frac{\lambda_2^2}{df_2}}$$

$\Rightarrow (N)^2$
normal distribution

i) MSR [Mean Square Regression] :-

→ This could also be called as avg Explained Variance per Independent feature.

$$MSR = \frac{ESS}{K}$$

independent var

ii) MSE [Mean Square Error] :-

This could also be called as average unexplained variance per degree of freedom.

$$MSE = \frac{RSS}{n - K - 1}$$

Calculate the F-statistic

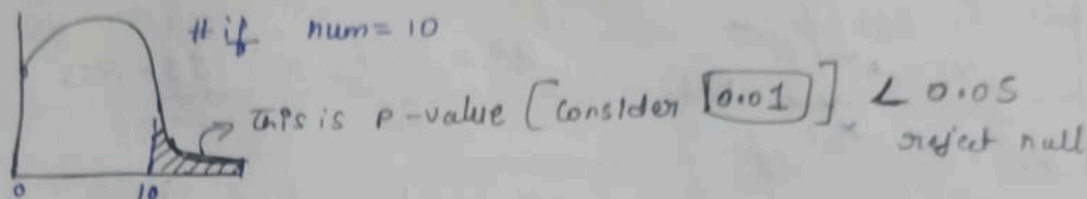
$$F\text{-statistic} = \frac{MSR}{MSE}$$

$$F\text{-statistic} = \frac{\frac{TSS - RSS}{k}}{\frac{RSS}{n-k-1}} \rightarrow \frac{\text{avg. is number}}{\text{f-dist}} \text{ which falls}$$

Step 6:- Determine the p-value

Compute the p-value associated with the calculated F-statistic using the F-distribution.

→ Plot the num obtained from f-statistics in f-dist



Step 7:- Compare the calculated F-statistic of the p-value to the chosen significance level α [in default = 0.05]

→ If the p-value $< \alpha$, reject the null hypothesis

→ { This indicates atleast 1 independent variable contributes in the variation/prediction.

→ If the p-value $\geq \alpha$, fail to reject the null hypothesis

→ This suggests that none of the independent variable contributes significantly to the prediction.

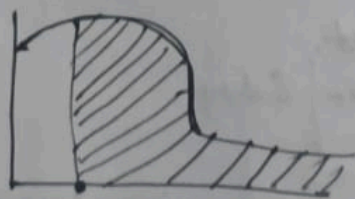
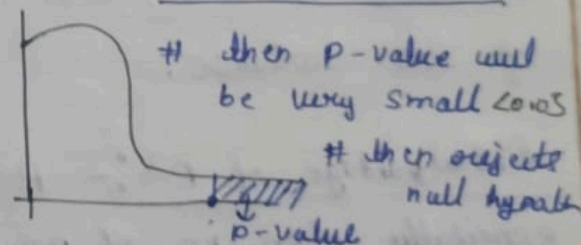
Intuition:- $F\text{-stat} = \frac{ESS}{k}$

avg. unexplained variance $\left\{ \frac{RSS}{n-k-1} \right\}$

} avg Explained Variance
if this avg large then
F-statistic is very large

if this avg is very large
F-statistic is very small

then p-value will be larger than
significance level (0.05)



fail to reject the null
hypothesis test

This indicates there is no
linear relationship b/w X & Y

Adjusted R-Squared

→ R^2 tells how well the model explains the target variable. But in multiple regression, if we keep adding more features (inputs), R^2 always increases even when the feature is useless.

→ This can cause overfitting — the model looks good but actually learns noise.

→ So adjusted R^2 solves this problem by penalizing extra useless features by reducing R^2 .

→ wt adjusted R^2 does:-

- It increase only if new input features actually improve the model.
- It decreases when useless features are added.

→ So it a better metric for models with many features.

$$\text{Adjusted } R^2 = 1 - \left[\frac{(1-R)^2 \times (n-1)}{(n-k-1)} \right]$$

$n \Rightarrow$ observations or rows
$k \Rightarrow$ input features

→ The denominator $(n-k-1)$:-

- Penalizes the model if you add more input variables
- when k increases, denominator reduced $\rightarrow$ making R^2_{adj} smaller

→ The numerator $(n-1)$:-

Related to dataset size $\rightarrow$ if dataset is small, penalty is bigger

Note: calculate both R-Squared & adjusted R^2

if the difference is very large then we get to know that, some irrelevant features are there. $\$$

T-statistic :-

Performing t-test for a simple linear regression, including the intercept term and using the p-value approach

Step 1:- State the null and alternative hypotheses for the slope and intercept coefficients:

→ for the slope coefficient (β_1):-

• Null hypothesis (H_0): $\beta_1 = 0$

{ No relationship b/w the predictor variable (x) and the response variable (y) }

• Alternative hypothesis (H_1): $\beta_1 \neq 0$

{ A relationship exists b/w x & y }

→ for the intercept coefficient (β_0):-

• Null hypothesis (H_0): $\beta_0 = 0$

Regression line passes through the origin.

• Alternative hypothesis (H_1): $\beta_0 \neq 0$

The regression line does not pass through the origin

Step 2:- Estimate the slope & intercept coefficient (b_0 & b_1)

→ using sample data, calculate the slope (b_1) and intercept (b_0) for the regression model.

Step 3:- Calculate the standard errors for slope & intercept

$$SE(b_1) = \frac{\sqrt{\sum (y_i - \hat{y}_i)^2}}{\sqrt{(n-2) \sum (x_i - \bar{x})^2}}$$

$b_1 \rightarrow \beta_1$

$$SE(b_0) = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{(n-2)} \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum (x_i - \bar{x})^2} \right]}$$

$b_0 \rightarrow \beta_0$

Step 4:- Compute the t-statistics for the slope & intercept

$$\frac{\text{t-statistic}}{\text{p-value}} = \frac{b_1 - 0}{SE(b_1)}$$

$$\frac{\text{t-statistic}}{\text{p-value}} = \frac{b_0 - 0}{SE(b_0)}$$